Bitcoin Sentiment–Price Prediction Agent

Evaluation Report | CrowdWisdomTrading X Sentiment Price Prediction Agent | 23 October 2026

Status: Preliminary evaluation. Recent X-post collection and news integration remain incomplete.

1. Executive Summary

This project investigates whether social-media sentiment and Bitcoin market data can help predict the next day's closing price. A Random Forest regressor was evaluated using walk-forward validation and compared with a naive baseline that predicts the next close using the current close.

Key result: Random Forest did not outperform the naive baseline. Walk-forward MAE was $598.63 versus $404.38, and RMSE was $789.32 versus $591.72.

2. Project Objectives

- Process historical Bitcoin-related tweet sentiment.
- Combine sentiment features with historical Bitcoin prices.
- Engineer features for machine-learning prediction.
- Train and evaluate a Random Forest model using walk-forward validation.
- Visualize historical sentiment and price movements.
- Evaluate preliminary trading-strategy risk metrics.

3. Dataset Description

The existing sentiment dataset covers 1 January 2022 to 22 June 2023 and includes tweet text, dates, sentiment labels/scores, and engagement information. Historical BTC prices were aligned with daily sentiment features.

Limitation: The dataset does not cover the latest 180 days of X posts. Recent X-post collection and relevant news integration remain incomplete.

4. Features and Methodology

Features include BTC closing price, volume, daily tweet count, average sentiment, aggregated likes and reposts, sentiment-label counts, and one-day and three-day returns. The target is the next-day closing price (target_next_day). Expanding-window walk-forward validation produced 214 out-of-sample predictions.

5. Model Evaluation

Metric	Random Forest	Naive baseline
MAE	$598.63	$404.38
RMSE	$789.32	$591.72

Lower is better. Random Forest MAE is approximately 48.0% higher than baseline MAE; RMSE is approximately 33.4% higher. The naive baseline predicts the next close as the current close.

A separate chronological holdout reported MAE $1,234.17, RMSE $1,449.70, and R^2 0.4377 on 107 rows. These values are from a different split and should not be mixed with walk-forward results.

6. Preliminary Strategy Evaluation

Metric	Model strategy	Buy and hold
Sharpe ratio	1.249	2.447
Sortino ratio	0.936	3.148
Maximum drawdown	-14.90%	-18.70%

The model strategy had a smaller reported maximum drawdown but lower Sharpe and Sortino ratios. Verify signal timing, price-return alignment, fees, slippage, and execution assumptions. These results do not establish live trading profitability.

7. Historical Sentiment Analysis

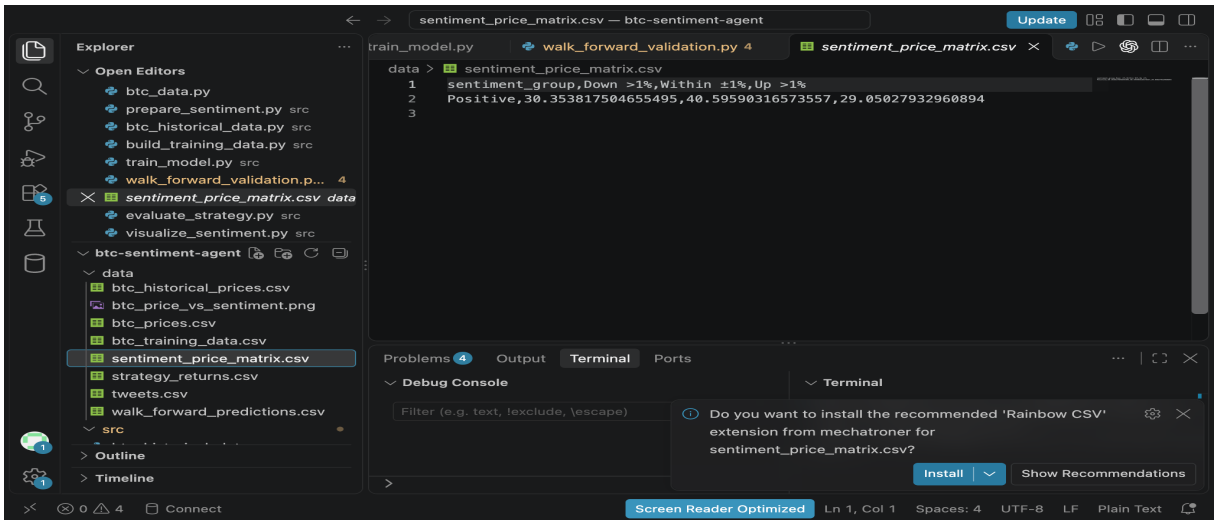
The chart compares BTC closing price with average daily tweet sentiment, approximately January 2022 through June 2023.



8. Sentiment–Price Matrix

Next-day return category (positive sentiment)	Percentage
Down more than 1%	30.35%
Within ±1%	40.60%
Up more than 1%	29.05%

These values describe the Positive sentiment row only and do not establish a reliable predictive relationship.



9. Key Findings

- Random Forest underperformed the naive baseline on both walk-forward error metrics.
- Positive sentiment was not consistently followed by an upward price move in the displayed matrix.
- The model strategy had a smaller reported maximum drawdown but weaker risk-adjusted performance than buy and hold.
- Historical sentiment data limits conclusions about current market conditions.
- The project has not established a reliable trading advantage.

10. Limitations

- Sentiment data covers 2022–2023 rather than the latest 180 days.
- Recent X-post collection and relevant news integration are incomplete.
- Social-media datasets may contain duplicates, spam, bots, and engagement bias.
- Model performance is weaker than the naive baseline in walk-forward validation.

- Strategy metrics require checks for transaction costs, slippage, and timing assumptions.
- No live trading validation has been performed.

11. Future Improvements

- Collect recent X posts after confirming data access and cost.
- Integrate relevant news and document sources and timestamps.
- Verify feature/target alignment to prevent leakage.
- Compare Random Forest with persistence and linear-regression baselines.
- Include transaction costs and slippage in backtests.
- Document setup steps and reproducible commands in the repository README.

12. Conclusion

The project demonstrates historical sentiment processing, feature engineering, machine-learning training, walk-forward validation, visualization, and preliminary strategy evaluation. The current Random Forest does not outperform the naive next-close baseline. Further validation is required before claiming forecasting skill or trading usefulness.

Disclaimer: Educational research only; not investment advice.